

Solving La Ghigliottina with a Typed PPMI Word Graph

Davide Carella¹

¹University of Bari Aldo Moro, Department of Computer Science, Master’s Degree in Computer Science (LM-18).

Abstract

La Ghigliottina, the final game of the Italian TV show *L’Eredità*, presents five clue words and asks for the single word that connects to all five. The connection is rarely plain semantic similarity: it is collocational and cultural, the solution forming a common multi-word expression with each clue. We build a *typed*, weighted word graph from the PAISÀ corpus, where edge types encode MWE-bearing syntactic patterns (noun–adjective, *X e Y*, *X di Y*, *X da Y*) and edge weights are pattern-conditioned PPMI, and we fold in curated IDIOM edges mined from Italian *polirematiche* and proverb lists. A simple flat PPMI-sum solver over this graph reaches **accuracy@1 0.31 / MRR 0.38** on the EVALITA 2018 NLP4FUN test set. We then document an extensive set of extensions that were explored and rejected: knowledge-graph embeddings, personalised PageRank, three additional corpora, compound / embedding-similarity / Wiktionary / ConceptNet edges, breadth-aware scorers, and a learned re-ranker. Every one of them improves recall or MRR but none improves accuracy@1: the task is bottlenecked by register-matched recall, not by ranking sophistication.

Keywords

La Ghigliottina, language games, PPMI, collocations, multi-word expressions, EVALITA NLP4FUN

1. Introduction and Motivations

La Ghigliottina gives the player five clue words and asks for the one solution word that links to all five simultaneously. The linking relation is almost never distributional similarity; it is *collocational and cultural*. The solution typically forms a fixed multi-word expression (MWE), a compound, or an idiom with each clue: *capo+stazione* → *capostazione* *sale e pepe*; *acqua di mare*; *sale grosso*.

The signal lives inside specific syntactic patterns rather than in a similarity space. Because of this purely distributional models (word2vec/fastText) are a poor structural fit. For example, a concrete test game like this:

legno, forte, continua, malattia, veloce → cassa

would not be solved by a similarity-based embedding model.

The design question this project answers is therefore *not* “which model is most powerful” but “which *signal* does the game actually rely on, and what is the smallest system that captures it directly.” The answer, established across many experiments, is that a plain typed co-occurrence graph enriched with curated idiom edges captures it, and that the residual error is a *recall* problem, missing clue → answer links, that no smarter scorer can close.

This report describes the shipped system (Section 3), evaluates it (Section 4), and, because the negative results are the substantive contribution, documents in Section 5 the full set of extensions that were tried and did not move the top-1 metric. None of the Section 5 machinery is part of the final codebase; it is retained here as the experimental record that justifies the minimal design.

2. Related Work

The EVALITA NLP4FUN shared task [1] framed *La Ghigliottina* as a computational language game. The seminal artificial player, OTTHO [2], solved it by a *knowledge-infusion* process over a dictionary, proverb lists, and Wikipedia: retrieval / co-occurrence over large Italian text and lexical resources, an approach that outperformed learned-embedding methods, the same qualitative outcome we reproduce here. Our contribution is not a new model but a controlled ablation showing *why*: on a graph of 30k entities, knowledge-graph embeddings are dominated by hubness, and every recall-broadening extension trades top-1 accuracy for rank-tail recall.

The strongest published player on this exact task is UNIOR4NLP, later released as *Il Mago della Ghigliottina* [3], [4], which won the EVALITA 2018 NLP4FUN evaluation [1] and its 2020 successor Ghigliottin-AI [5], reporting **61.0% accuracy on the 2018 test set and 68.6% on the 2020 set**. This project set out to *replicate* that system and then extend it: its MWE-based design is the starting point we deliberately mirror, sharing PAISÀ, the De Mauro *polirematiche* (30,633 MWE entries) and *vocabolario di base*, curated proverb lists, X+Y compounds, and essentially the same six MWE-bearing patterns (noun–adjective, X e Y, X di Y, X (prep) (det) Y, compounds) that we encode as typed edges. The extensions in Section 5, knowledge-graph embeddings above all, were meant to push *past* their number. Neither goal succeeded: our reproduction on the same 2018 data reaches only 0.31, and none of the extensions moved acc@1. The shortfall is a coverage gap, not a scoring one, their advantage comes from broader curated resources (itWaC, 1.5B words; Wikipedia titles; morphologically expanded De Mauro entries; far larger proverb lists), which is precisely the *register-matched associative phrase resource* our error analysis (Section 5) isolates as the one axis with demonstrated acc@1 headroom. Falling short of their result is, in that sense, the same finding read from the other side: on this game, coverage, not ranking, is the binding constraint.

Methodologically the system rests on pointwise mutual information (PMI) and its positive variant (PPMI), the standard association measure for collocations, applied *within* each syntactic relation rather than over a single undifferentiated co-occurrence window. Because the game’s signal is syntagmatic rather than paradigmatic, purely distributional embeddings, word2vec [6] and fastText [7], [8], are a poor structural fit, a hypothesis the knowledge-graph-embedding experiments (Section 5) confirm.

3. Proposed Approach

3.1. Description of the solution and dataset

The shipped solver has three stages.

1. *Build a typed, weighted word graph* from the PAISÀ corpus. Edge types are MWE-bearing relations extracted by adjacency patterns over lemma+POS sequences: MOD (noun-adjective, either order), BINOMIAL (X e Y), DI (X di Y), and DA (X da Y). Edge weights are pattern-conditioned PPMI: PMI is computed within each relation’s own pair population, so the association of a pair is measured against the right baseline rather than a global one.
2. *Fold in curated IDIOM edges* mined from Italian *polirematiche* and proverb lists. These supply the clue → answer links that running-text adjacency patterns structurally cannot produce (idioms are phrase lists, not local syntactic patterns). Each curated edge is added with a constant weight (default 5.5, the single tuned scalar of the system; see Section 4.3).
3. *Solve with flat PPMI-sum*. For each clue, take the strongest edge to a candidate (a max over relations and directions, computed once as an adjacency map), then *sum* those best-edge weights across the five clues. The candidate with the highest sum is the predicted solution.

Dataset. Evaluation uses the EVALITA 2018 NLP4FUN test set: 100 games in JSON-Lines, each with `hint1..hint5` and a `gold sol`. This set is used for scoring only, never for training, the solver has no learned parameters.

3.2. Main technical details

Corpus. The base graph is built from the full PAISÀ corpus [9] (annotated CoNLL, CC BY-NC-SA), streamed rather than materialised. The loader (`corpus.py`) reads `gzip/xz` transparently, maps both the TanI and Universal Dependencies POS tagsets to a coarse common set, falls back from a missing lemma to the surface form, and yields one sentence at a time so that memory is bounded by vocabulary size, not corpus size.

Vocabulary. A frequency list defines the candidate vocabulary: the top $\approx 50,000$ lemmas with frequency ≥ 5 . Only pairs whose *both* endpoints are in this vocabulary are counted.

Pair extraction and PPMI. For each sentence the extractor walks tokens and emits: a MOD pair for adjacent NOUN–ADJ; and, for tokens two apart joined by the right function word, a BINOMIAL / DI / DA pair keyed on the connective lemma and POS. For each relation, the weight of its edge follows these computations

$$\begin{aligned}
p(h, t \mid r) &= \frac{c(h, t, r)}{\text{tot}(r)} \\
p(h \mid r) &= \sum_{t'} \frac{c(h, t', r)}{\text{tot}(r)} \\
p(t \mid r) &= \sum_{h'} \frac{c(h', t, r)}{\text{tot}(r)} \\
\text{pmi}(h, t \mid r) &= \ln \left(\frac{p(h, t \mid r)}{p(h \mid r)p(t \mid r)} \right) = \\
&= \ln \left(\frac{c(h, t, r) \cdot \text{tot}(r)}{\sum_{t'} c(h, t', r) \cdot \sum_{h'} c(h', t, r)} \right) \\
\text{ppmi}(h, t \mid r) &= \max(0, \text{pmi}(h, t \mid r))
\end{aligned}$$

where h is the head lemma, t the tail lemma, r the relation, and $c(h, t, r)$ is the count of the triple. The total count $\text{tot}(r)$ is the sum of all counts for that relation.

Each weight is kept only when $c(h, t, r) \geq 5$ (`min_count`). The curated idiom edges are then merged in as a separate IDIOM relation at constant weight.

Curated idiom mining. Three Italian resources, 31k De Mauro *polirematiche*, 4.7k tagged *polirematiche*, and 369 proverbs, are tagged with spaCy [10] (`it_core_news_sm`); only *nominal* content lemmas (NOUN/ADJ/PROPN) that are in the vocabulary are kept, and an undirected edge is emitted for every unordered lemma pair co-occurring in a phrase. Restricting to nominal lemmas is deliberate: including VERB/ADV pulls in hub lemmas (*fare, avere, essere*) that pollute the sum scorer. The 36k phrases yield $\approx 18.5\text{k}$ distinct nominal pairs.

3.3. Optional solution explanation

Beyond predicting the solution, the system can optionally *explain* it. Given the predicted answer and the five clues, a small instruction-tuned LLM, Qwen3-1.7B [11] run locally through the Hugging Face transformers pipeline, is prompted in Italian to produce one sentence per clue naming the multi-word expression, compound, or idiom that links it to the answer (greedy decoding, thinking mode disabled). This is a post-hoc presentation layer, exposed via `--describe`; it plays no part in solving and does not affect the ranking metrics.

4. Evaluation

4.1. Setup and metrics

The graph is built from full PAISÀ (599,964 typed triples over 32,018 entities at `min_count=5`, top-50k vocabulary). We report, over the 100-game EVALITA test set: *found* (gold present in the top 100), MRR, and `accuracy@{1,5,10,100}`. The primary metric for the game is `accuracy@1`.

4.2. Main result

CONFIGURATION	FOUND	MRR	ACC@1	ACC@5	ACC@10	ACC@100
flat-PPMI(sum), PAISÀ only	60	0.2881	0.240	0.330	0.380	0.600
flat-PPMI(sum) + IDIOM (w=5.5)	67	0.3841	0.310	0.480	0.520	0.670

Table 1: The shipped configuration against the plain baseline. Curated idiom edges are the single largest lever in the project.

Curated idiom edges lift `accuracy@1` from 0.240 to **0.310** (+29% relative) and MRR from 0.288 to 0.384. The idiom weight (5.5) is the system’s only tuned scalar, selected on the *train* split (Section 4.3), and it sits on a wide inverted-U plateau rather than a knife-edge fit. A constant weight beats scaling by how many phrases share a pair: the game draws on rare expressions, so uniform trust wins.

4.3. Idiom weight tuning

The constant IDIOM edge weight is the only free scalar in the system, and it is tuned on the *train* split (965 EVALITA games), never on the 100-game test set used for the results above. The sweep is run by `tools/idiom_tune.py`, which rebuilds the merged graph at each weight and evaluates the flat PPMI-sum solver on train, parallelising the 21 weights

across worker processes. Accuracy@1 traces a clean inverted-U, rising from 0.167 at $w = 0$ (idioms off) to a peak of **0.263 at $w = 5.5$** and decaying gently thereafter as over-weighted flat idiom edges start to outrank genuine PPMI collocations.

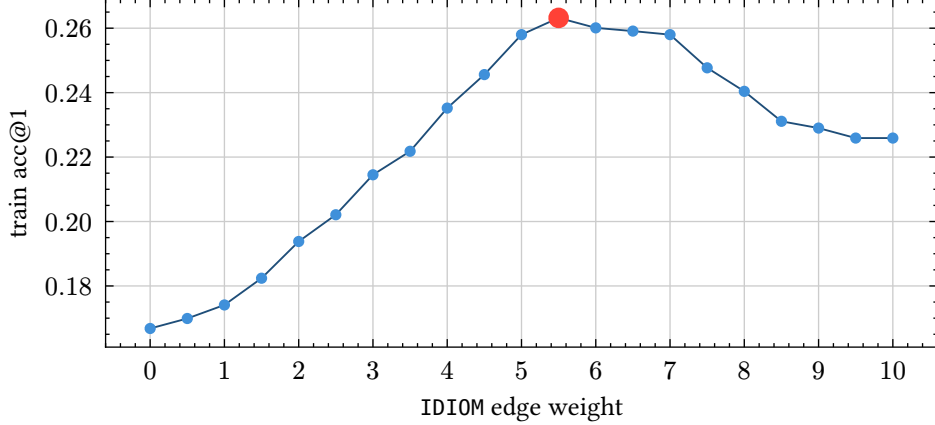


Figure 1: Train-set accuracy@1 as a function of the constant IDIOM edge weight (tools/idiom_tune.py, 965 train games, PAISÀ+IDIOM graph). The peak (marked) is at $w = 5.5$, on a broad plateau spanning $w = 5-7$.

The plateau is wide ($w = 5-7$ all stay within 0.005 of the peak), so the exact value is not delicate; 5.5 is adopted as the shipped default. The absolute numbers are lower than the test-set results because the train split is larger and harder, but the tuning is what matters: the weight is fixed from train and carried over unchanged to test.

4.4. Description quality

The optional explanation pass (Section 3.3) is scored against the gold desc field of the 100 test games. Each explanation is generated for the solver’s own top-1 prediction, so the numbers fold in solver errors as well as phrasing differences.

BLEU	ROUGE-1	ROUGE-2	ROUGE-L	ROUGE-LSUM	BERTSCORE
1.70	0.248	0.035	0.177	0.170	0.658

Table 2: Explanation quality of the Qwen3-1.7B - -describe pass against the gold desc field (100 test games). BLEU is on the sacreBLEU 0–100 scale; the rest are 0–1.

The low n-gram overlap (BLEU 1.70, ROUGE-L 0.18) is expected: the explanations are free-form generative Italian that rarely reuses the gold wording, and roughly two thirds of them describe a predicted word that differs from the gold (acc@1 0.31). *BERTScore* 0.66, which measures contextual-embedding similarity rather than surface overlap, confirms the explanations are semantically on-topic.

5. Explored Extensions and Negative Results

This section documents the extensions that were built, evaluated, and then left out of the final system. They are the experimental core of the project: they establish that accuracy@1 0.31 is a genuine ceiling for this signal, and that the residual error is a recall problem. The recurring pattern is stated once and confirmed repeatedly: *every lever improves recall / MRR / acc@{10,100} but not acc@1*.

5.1. Knowledge-graph embeddings (the original thesis) – decisive loss

The project’s original hypothesis was that learned KG embeddings could score *unobserved* clue→solution links that a graph baseline cannot. Trained with PyKEEN [12] over the same typed triples, ComplEx [13] and RotatE [14] lost badly.

METHOD	FOUND	MRR	ACC@1	ACC@5	ACC@10	ACC@100
flat-PPMI(sum)	60	0.2881	0.240	0.330	0.380	0.600
PageRank (personalised)	46	0.0829	0.050	0.110	0.160	0.460
KGE ComplEx (full, 300 ep)	21	0.0230	0.010	0.030	0.040	0.210
KGE RotatE (full, 150 ep, 16 negs)	37	0.0676	0.040	0.080	0.100	0.370

Table 3: KGE and PageRank vs. the flat baseline (PAISÀ only, all candidates).

A head-ranking sanity check explains the collapse: for known MWE partners, RotatE ranks the true head at #66–170. Over 30k entities, link prediction is dominated by hubness, frequent adjectives and prepositions absorb probability mass so the model smears its score instead of pinning the collocates. This reproduces the EVALITA finding that co-occurrence methods beat learned embeddings on this game.

5.2. Candidate restriction and breadth-aware scorers

Noun-restriction (restrict candidates to noun-dominant lemmas) hurts: MRR 0.288 $\rightarrow$ 0.236, found 60 $\rightarrow$ 55. The sum scorer already ranks the true noun above adjective distractors, so the restriction only drops recall.

The game’s rule is that the answer connects to all five clues, which motivates scorers that reward breadth over a single strong edge. Both hurt top-1:

SCORER	FOUND	MRR	ACC@1	ACC@5	ACC@10	ACC@100
sum (champion)	67	0.3841	0.310	0.480	0.520	0.670
coverage (count clues covered)	65	0.2867	0.200	0.400	0.500	0.650
logsum (diminishing returns)	68	0.3783	0.290	0.480	0.570	0.680

Table 4: Breadth-aware scorers on the PAISÀ+IDIOM champion graph.

PPMI magnitude pins top-1; rewarding breadth dilutes it. Coverage-first is worst (0.310 $\rightarrow$ 0.200): edge existence is noisy (hub adjectives, flat-weight idiom edges all count as “covered”), so weak-but-complete junk outranks strong-but-partial truth. logsum softens this but still trades acc@1 for recall. (sum is the shipped default; the alternatives remain selectable via --combine but are not recommended.)

5.3. Additional corpora – register beats volume

Three further corpora were joined into the PPMI scan. The lesson is that *register*, not volume, is the lever, and that candidate vocabulary must stay PAISÀ-defined while an extra corpus only enriches counts.

ADDED CORPUS	+TOKENS	FOUND	MRR	ACC@1	ACC@10	ACC@100	VERDICT
– (PAISÀ only)	–	60	0.2881	0.240	0.380	0.600	baseline
Wikipedia (CoNLL’17)	+650M	61	0.2854	0.240	0.410	0.610	recall only
OpenSubtitles 12M	+85M	63	0.3193	0.270	0.420	0.630	helps
OpenSubtitles full	+766M	66	0.3164	0.250	0.450	0.660	↑ recall ↓ acc@1
itWaC (1 of 21 archives)	+93M	60	0.2785	0.220	–	0.600	hurts

Table 5: Corpus joins, flat-PPMI(sum) with PAISÀ-defined candidates.

OpenSubtitles [15] (12M-line subset) is the only corpus join that improves acc@1 (0.240 $\rightarrow$ 0.270): register-matched dialogue sharpens the game’s collocations even though it barely grows the graph. *Wikipedia* (CoNLL 2017 UD parse) doubled the graph yet gave recall only because of its encyclopedic register. *itWaC* [16] (general web text, same register as PAISÀ) slightly hurt and the run was stopped after one archive. Scaling OpenSubtitles to the full 766M tokens broadened recall but lowered acc@1 to 0.250, more of the right register is not monotonically better, because extra collocations also reinforce distractors at rank 1. Notably, none of these joins beats the curated idiom edges, and stacking OpenSubtitles with idioms gives the best recall/MRR (found 70, acc@100 0.70) while acc@1 (0.32) stays within noise of pure PAISÀ+IDIOM (0.31).

5.4. Curated compound edges – small recall gain, acc@1 held

The headline *capostazione* case is a single-token compound invisible to the adjacency patterns. A heuristic decomposer (a node splits unambiguously into two other frequent nominal nodes) produced 1,708 COMPOUND edges. On top of the champion this lifts found 67 $\rightarrow$ 69 and acc@100 0.670 $\rightarrow$ 0.690 with acc@1 steady at 0.310, a clean but small recall gain. Kept out of the final system as marginal.

5.5. The recall-miss anatomy – why the ceiling exists

A gold-rank diagnostic on the champion graph partitions the 100 test games: 31 already rank #1; 21 at rank 2–10; 17 at rank 11–100; 31 missing (>100 , recall-dead). Raising the vocabulary from 50k to 100k adds zero new gold answers on test (98/100 gold are already nodes), this means that the bottleneck is not node coverage. Dumping the 31 missing games shows the gold answer connects to 0 clues (10 games), 1 clue (18), or 2 clues (3): the sum scorer cannot lift

an answer with ≤ 2 weak edges over the flood that touch 3–5. The missing links are *syntagmatic collocations* *people simply know* (*cassa forte / continua / malattia, terza età, vita da cani*) plus a proper-noun residue. This is precisely why similarity and association-based edges cannot help: the gap is associative clue→answer *phrases*, not synonyms.

5.6. Off-corpus edge sources – all the right idea, all too thin

Four external resources were folded in as new edge relations to attack the 31 recall-dead games. Each is qualitatively the right kind of signal and quantitatively insufficient.

SOURCE	EDGES	FOUND	MRR	ACC@1	DECISIVE RESULT
<i>baseline</i>	–	67–69	0.3841	0.310	–
SIM (fastText cc.it.300)	≈268k	67–68	0.3993	0.350	train found 629→629
Wiktionary MWEs (kaikki)	21,358	69	0.3931	0.310	recovered 2 of 31; gate ≥ 3 fails
ConceptNet 5.7 (IT)	24,854	67	0.3841	0.310	train found 629→630

Table 6: Off-corpus edge sources on the PAISÀ+IDIOM champion (test unless noted). The decisive number for a recall lever is *found*.

SIM (fastText [8] cosine neighbours) leaves found unchanged: nearest neighbours are *paradigmatic* (synonyms of already-connected words), not the syntagmatic clue→answer collocations needed, so they re-rank within reach instead of extending it. It must be weighted below PPMI ($w \leq 3$) or it floods every clue with synonyms and demotes real answers.

Wiktionary MWEs (via wiktextract [17]) recovered exactly 2 of the 31 misses (13.9k phrases collapse to only ≈2.9k net-new pairs after overlap with De Mauro), failing a pre-registered “ ≥ 3 recovered” gate.

ConceptNet [18], an association KB by construction, bought exactly one net-new reachable answer: Italian single-lemma coverage is too sparse.

5.7. Learned re-ranker – the ceiling is not a ranking artefact

The 18 mid-rank (2–10) games look like a re-ranking opportunity, and *train.json* (965 games) had so far been used only to fit the single idiom-weight scalar (Section 4.3). We replaced the scalar sum with a 10-feature linear scorer (sum_ppmi, coverage, max_edge, logsum, and per-relation coverage) and fit the weights on train and evaluated on test.

CONFIGURATION	FOUND	MRR	ACC@1	ACC@5	ACC@10	ACC@100
sum (baseline), test	69	0.3841	0.310	0.480	0.520	0.690
reranker, train (965)	–	–	0.350	–	–	–
reranker, test	70	0.4043	0.330	0.490	0.580	0.700

Table 7: Learned re-ranker (coordinate ascent on train acc@1) vs. the sum baseline.

The learned weights reach 0.350 on train but only 0.330 on test, a two-game wobble over the sum baseline (0.310) that is well within sampling noise on 100 games. An optimal linear combination of every available feature, fit on 965 games, delivers no robust top-1 gain on held-out data. The re-ranker does robustly help recall/MRR (the part that generalises), confirming once more that the bottleneck is the 31 recall-dead games, not the scorer.

6. Conclusions and Limitations

A minimal system – a typed, pattern-conditioned PPMI graph over PAISÀ, enriched with ≈18.5k curated idiom edges, scored by a flat PPMI-sum – reaches **accuracy@1 0.31 / MRR 0.38** on EVALITA 2018 NLP4FUN. It has no learned parameters and a single tuned scalar (the idiom weight, on a wide plateau). Against it we measured knowledge-graph embeddings, PageRank, noun-restriction, breadth-aware scorers, three corpus joins, compound / embedding-similarity / Wiktionary / ConceptNet edges, and a learned re-ranker. The result is unambiguous and repeated across roughly eight independent experiments: **every extension improves recall, MRR, and acc@{10,100}, but none improves accuracy@1**. The curated idiom edges, the purest register match, are the only lever that ever moved top-1.

Limitations.

1. The heuristic decompounder and the nominal-only idiom filter miss verb-initial compounds and verbal idioms.

2. The real ceiling is *recall*: 31 of 100 test games have gold answers that exist as graph nodes but lack the clue→ answer edges the corpus patterns cannot produce. Of these, ≈ 18 are dictionary-MWE gaps, ≈ 9 are cultural/proper-noun knowledge, and 2 are vocabulary gaps. Closing them requires a *register-matched, associative phrase resource* keyed on co-occurrence *presence*, not a smarter ranker, more generic text, or paradigmatic similarity, all of which were shown to be dead ends. That resource is the one direction with demonstrated acc@1 headroom and the natural next step.

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